

Lec11

Tailwind CSS

Tailwind CSS is a **utility-first CSS framework** that allows you to build modern and responsive user interfaces directly in your HTML by using predefined utility classes.

Unlike Bootstrap, which provides ready-made components (buttons, cards, navbars), Tailwind provides **small utility classes** that you combine to create your own designs.

Why use Tailwind CSS?

Faster UI development

Highly customizable

Responsive design support

Smaller CSS bundle (after optimization)

No need to write much custom CSS

Popular in modern frameworks like React, Next.js, and Vue

Include of Tailwind

There are several ways to include **Tailwind CSS** in your project.

1. Using CDN (Easiest for Learning)

Add this script inside the `<head>` section:

```
<head>
  <script src="https://cdn.tailwindcss.com"></script>
</head>
```

This is best for:

- Beginners
- Practice
- Small projects

Using npm (Recommended for React, Next.js, Vue)

Install Tailwind CSS

npm install tailwindcss @tailwindcss/postcss postcss

Configure Tailwind

Create a CSS file:

```
@import "tailwindcss";
```

Then import it into your project.

This method is recommended for:

- React
- Next.js
- Vue
- Production projects

Basic Tailwind Example

```
<!DOCTYPE html>
<html>
<head>
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="flex justify-center items-center h-screen">

  <button class="bg-blue-500 text-white px-6 py-2 rounded-lg hover:bg-blue-700">
    Click Me
  </button>

</body>
</html>
```

Bootstrap vs Tailwind

Feature	Bootstrap	Tailwind
Approach	Component-based	Utility-first
Customization	Moderate	Very High
Ready-made Components	Yes	No
Learning Curve	Easy	Moderate
CSS Writing	Less	Almost none
Design Flexibility	Limited	Excellent

Advantages of Tailwind CSS

Rapid development

Highly customizable

Responsive by default

Consistent styling

Works well with React and Next.js

No need for large custom CSS files

Disadvantages of Tailwind CSS

HTML can become lengthy with many classes

Initial learning curve

Not many built-in UI components compared to Bootstrap

Task

Build a Tailwind-based responsive webpage